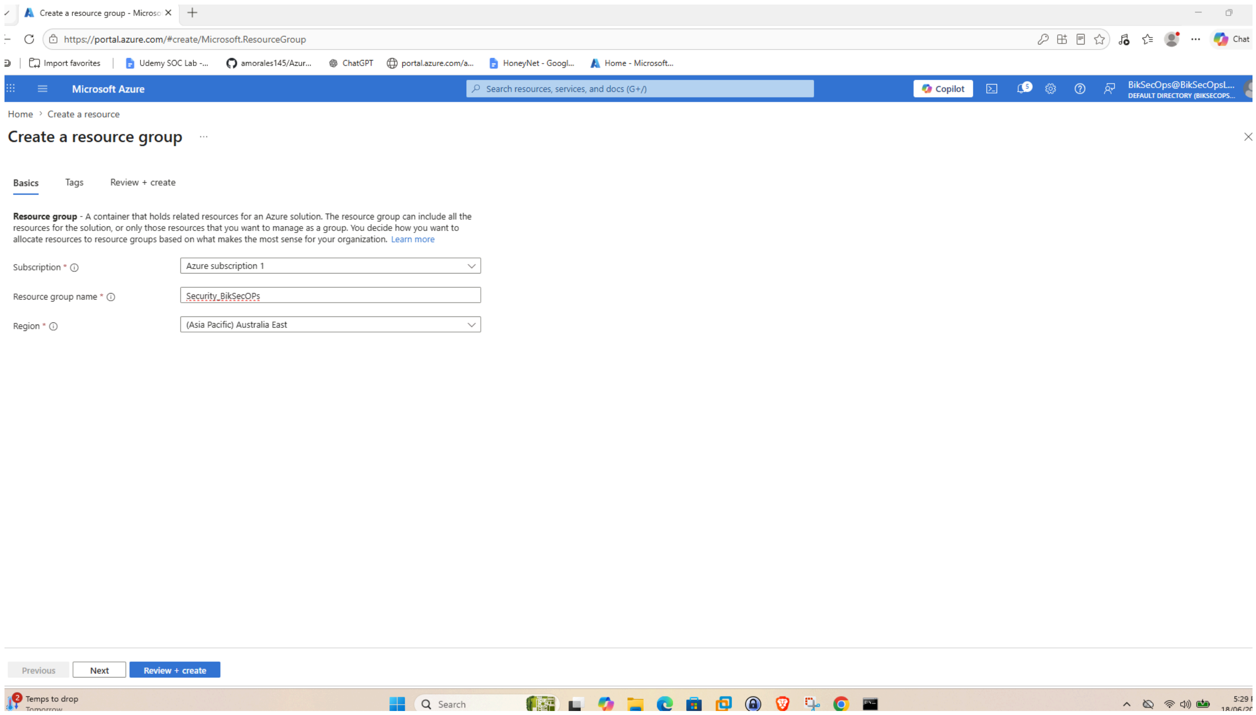
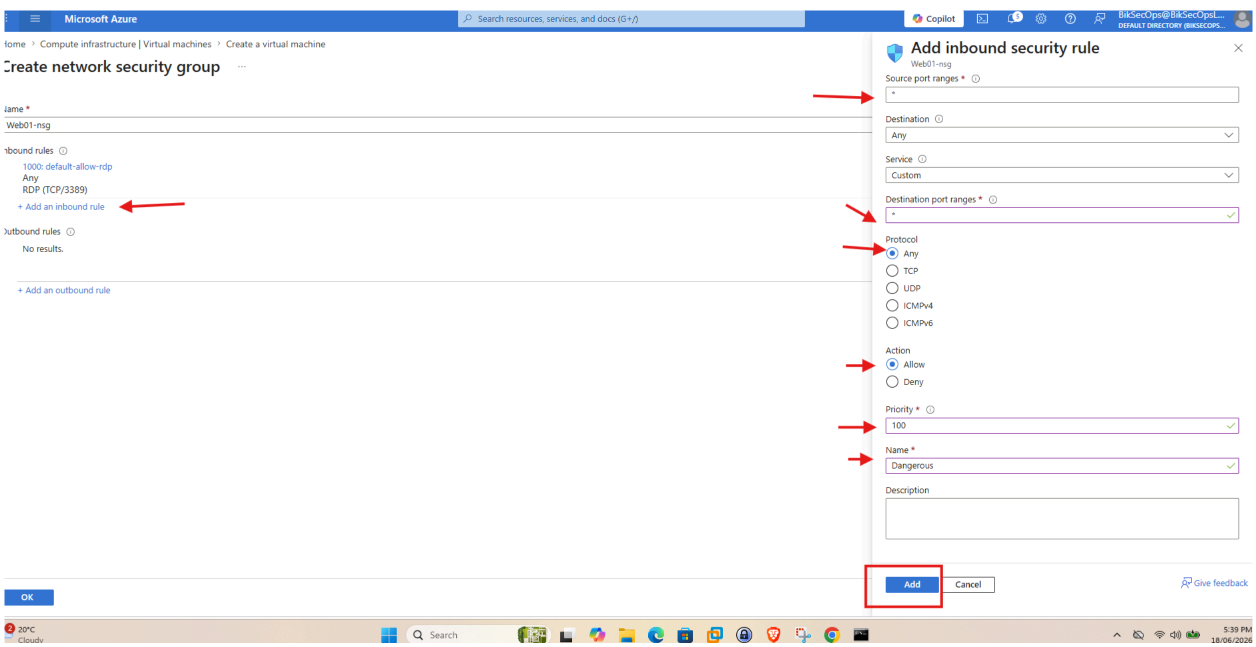
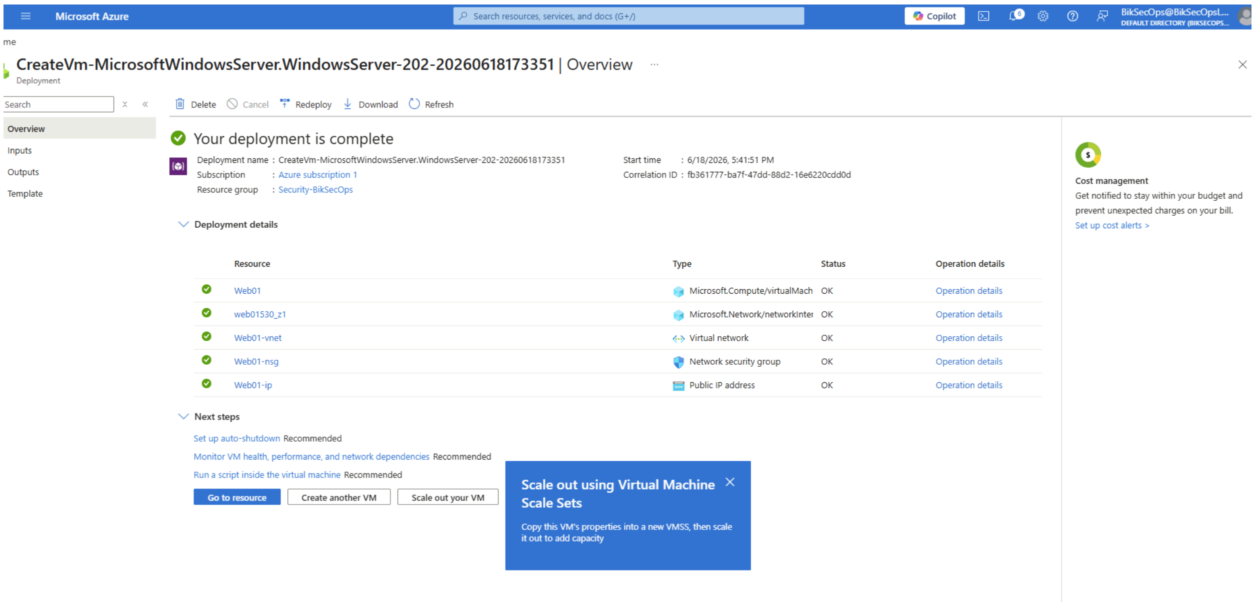
RG: Security-BikSecOps

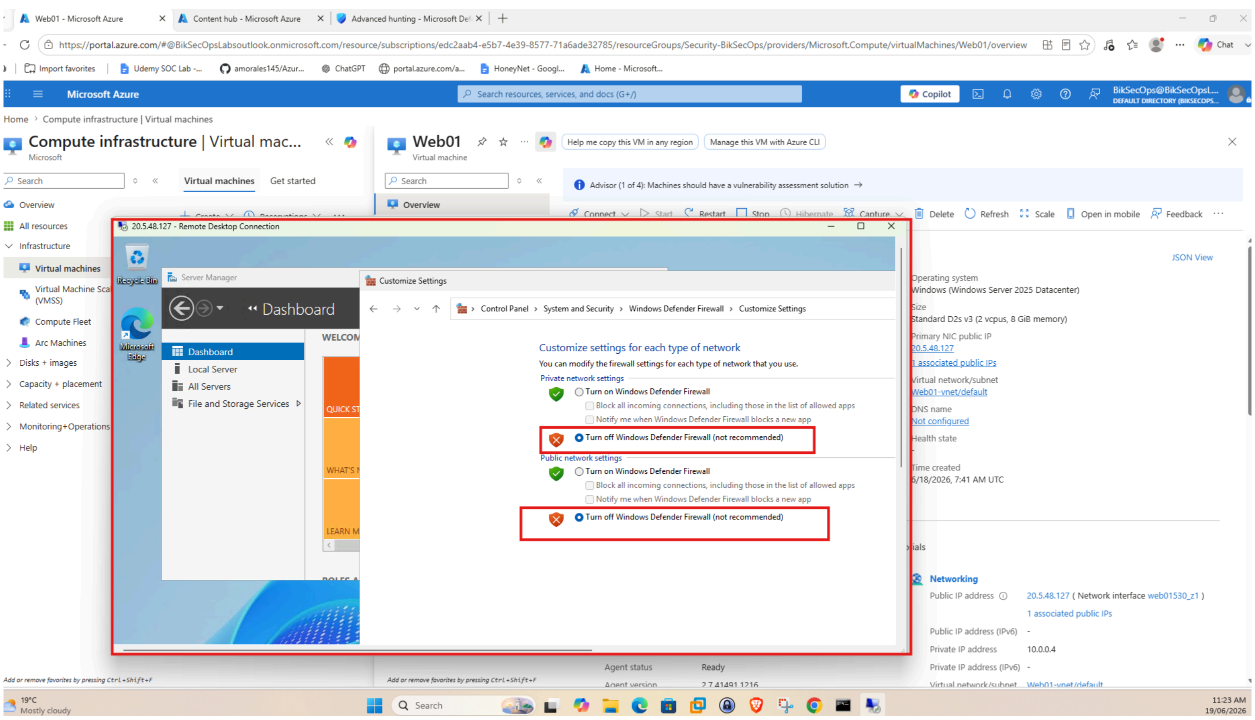


Create VM with new NSG inbound rule Any Any *:

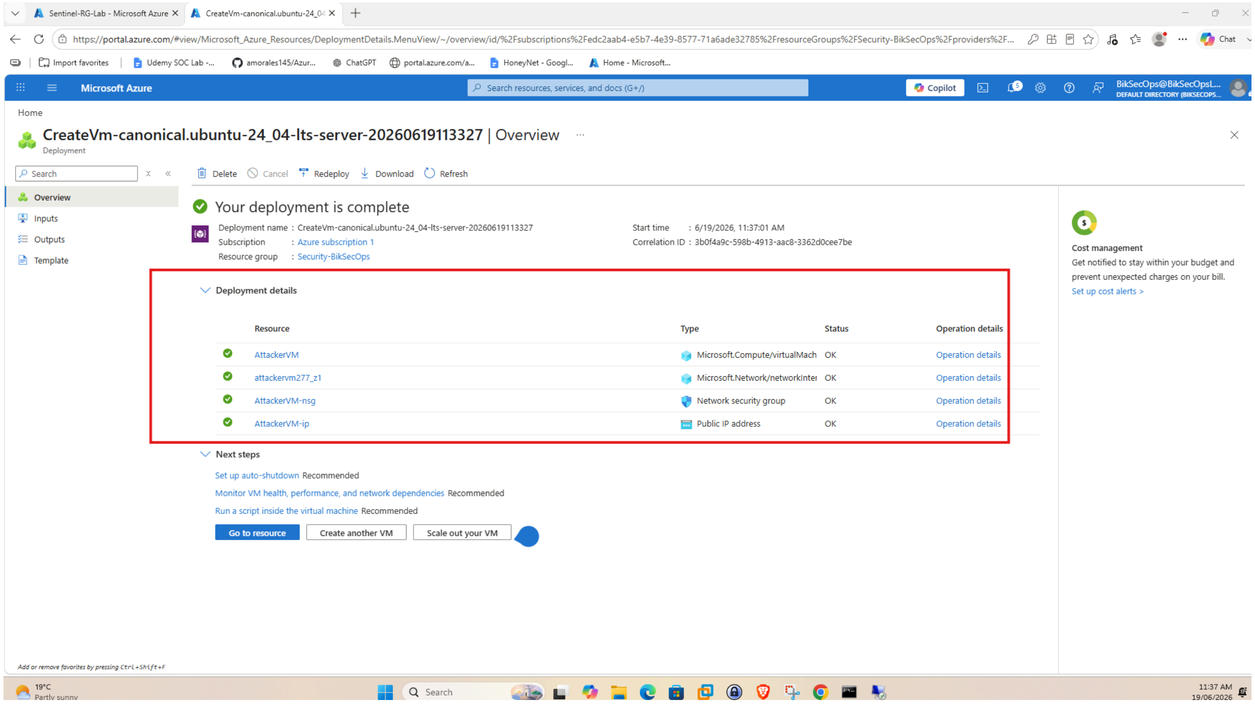




RDP and Turn off the host Fire wall on Vm windows Server:

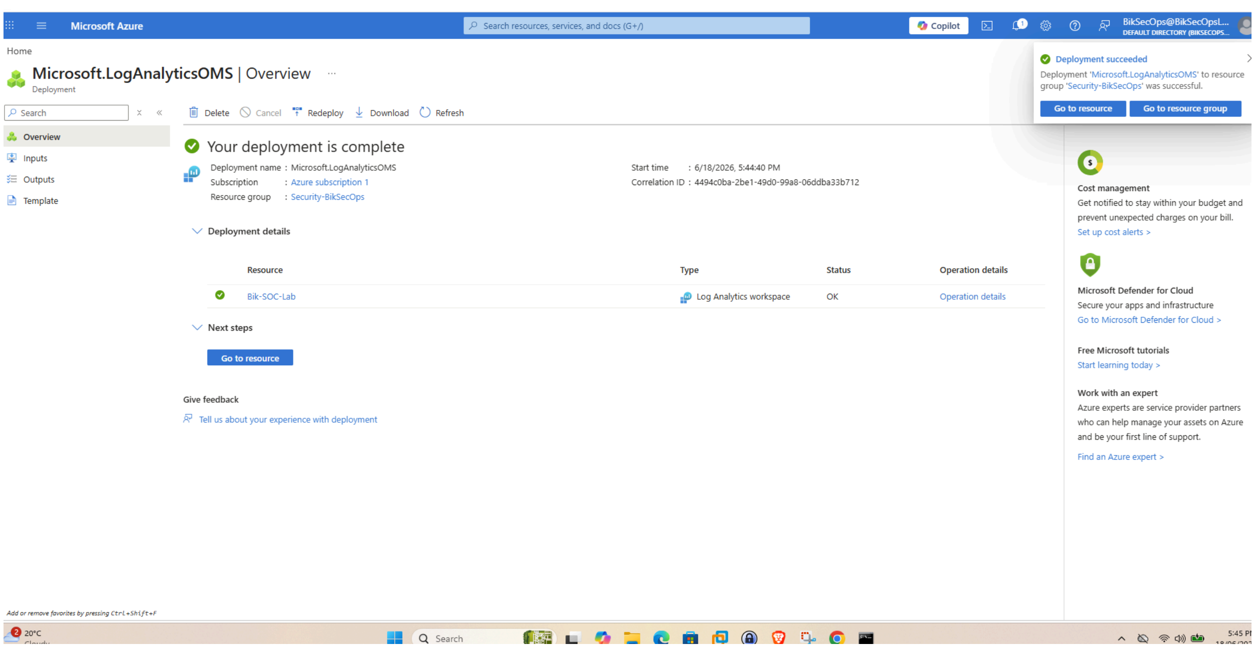


Create AttackerVM Ubuntu VM

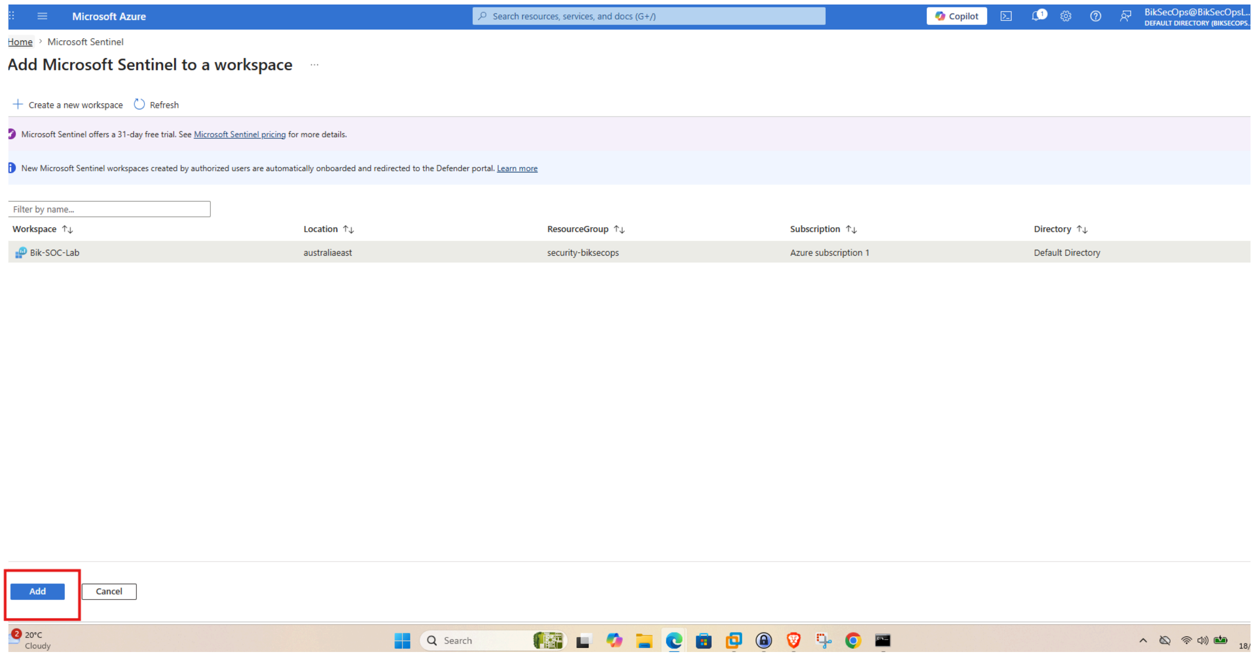


Create LAW:

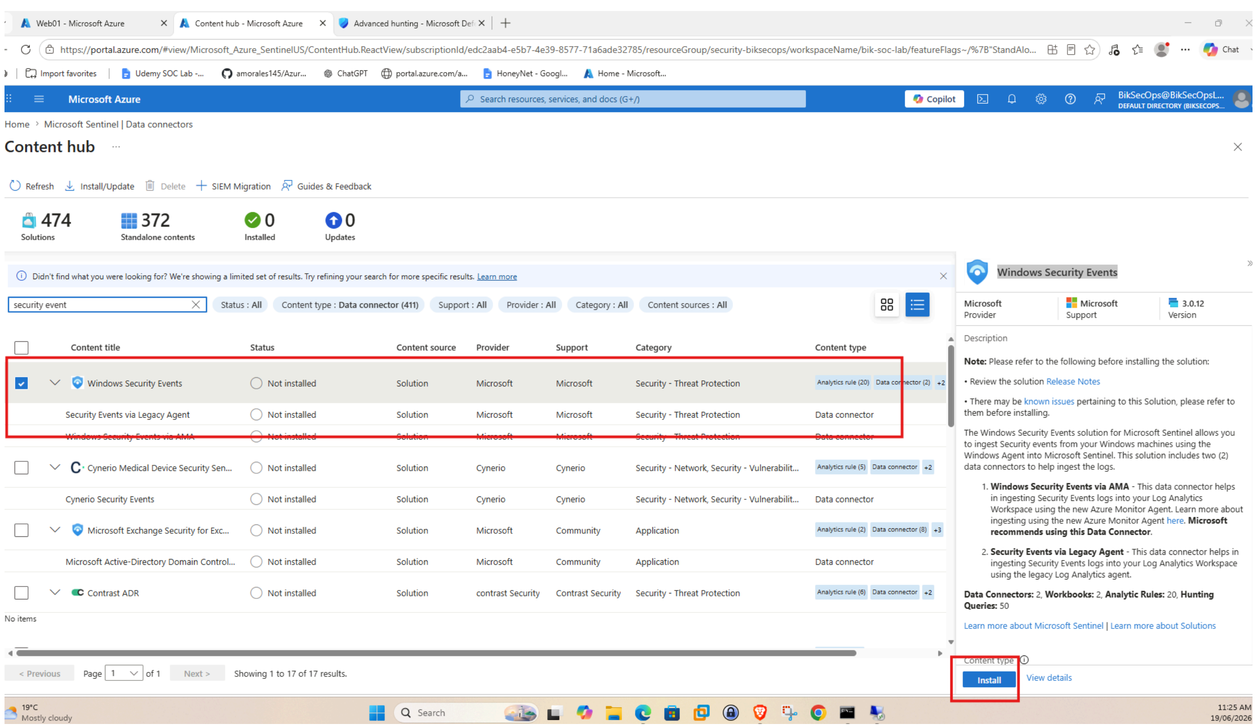
Bik-SOC-Lab



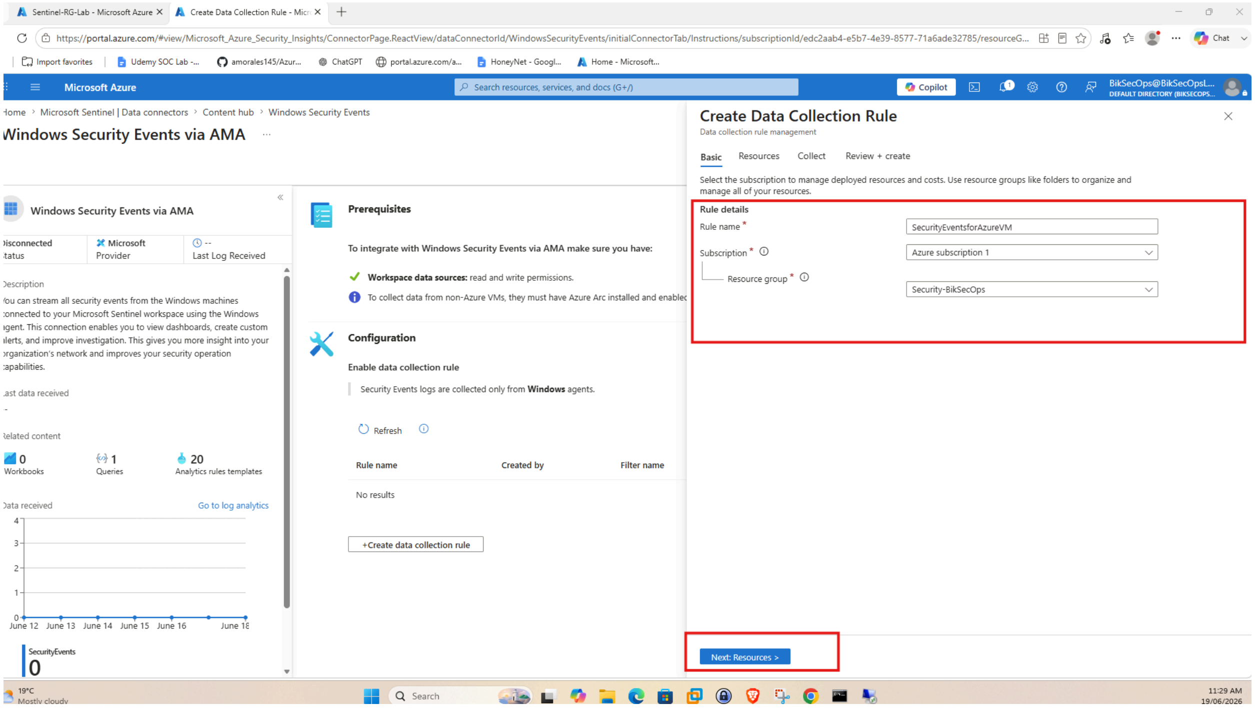
Add to sentinel

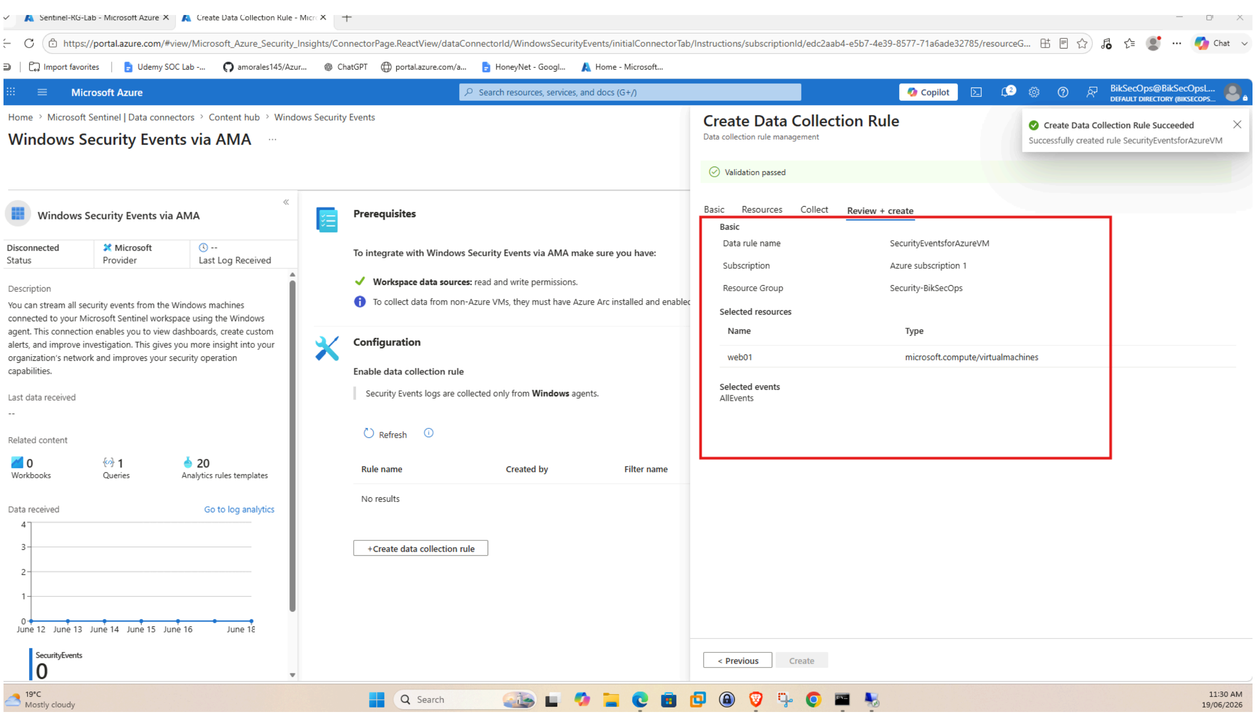


Install Windows Security Events AMA Connector to Sentinel:

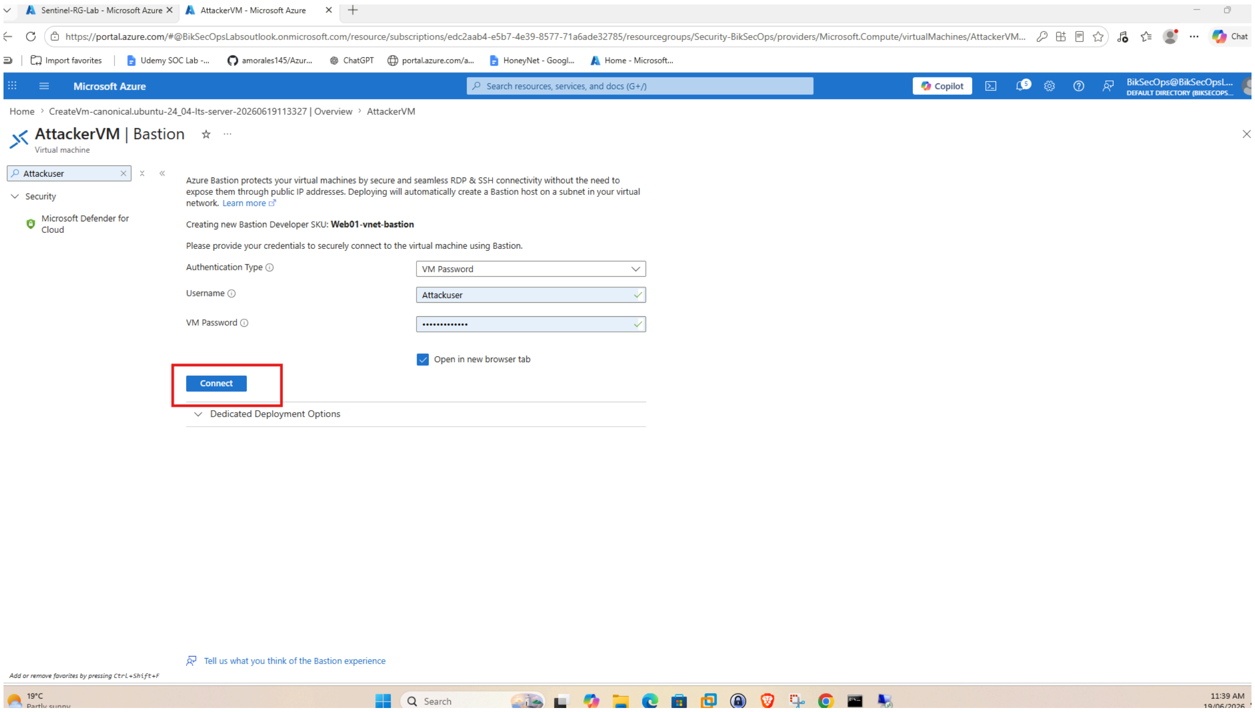


Configure : Windows Security Events AMA and create data collection rule:

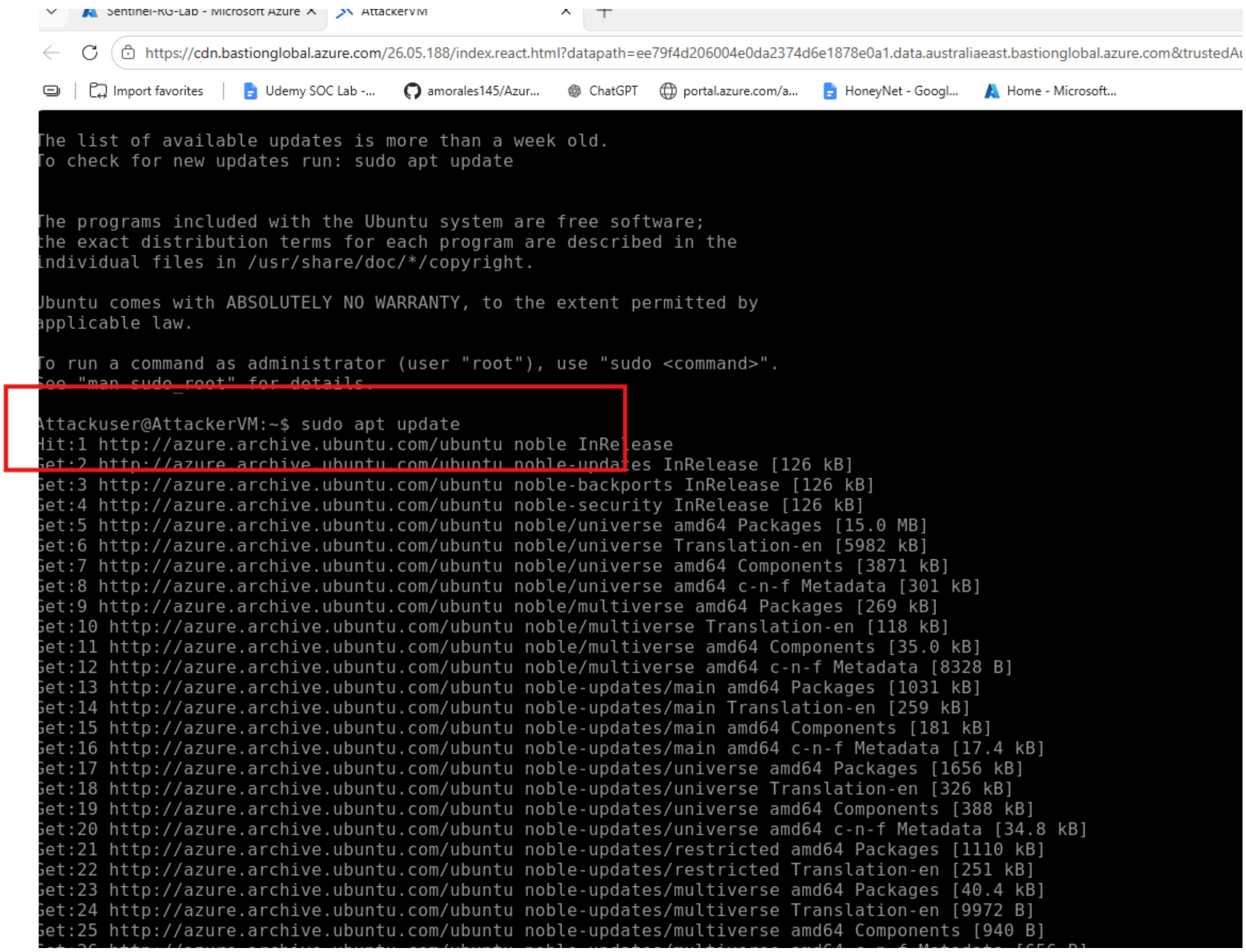


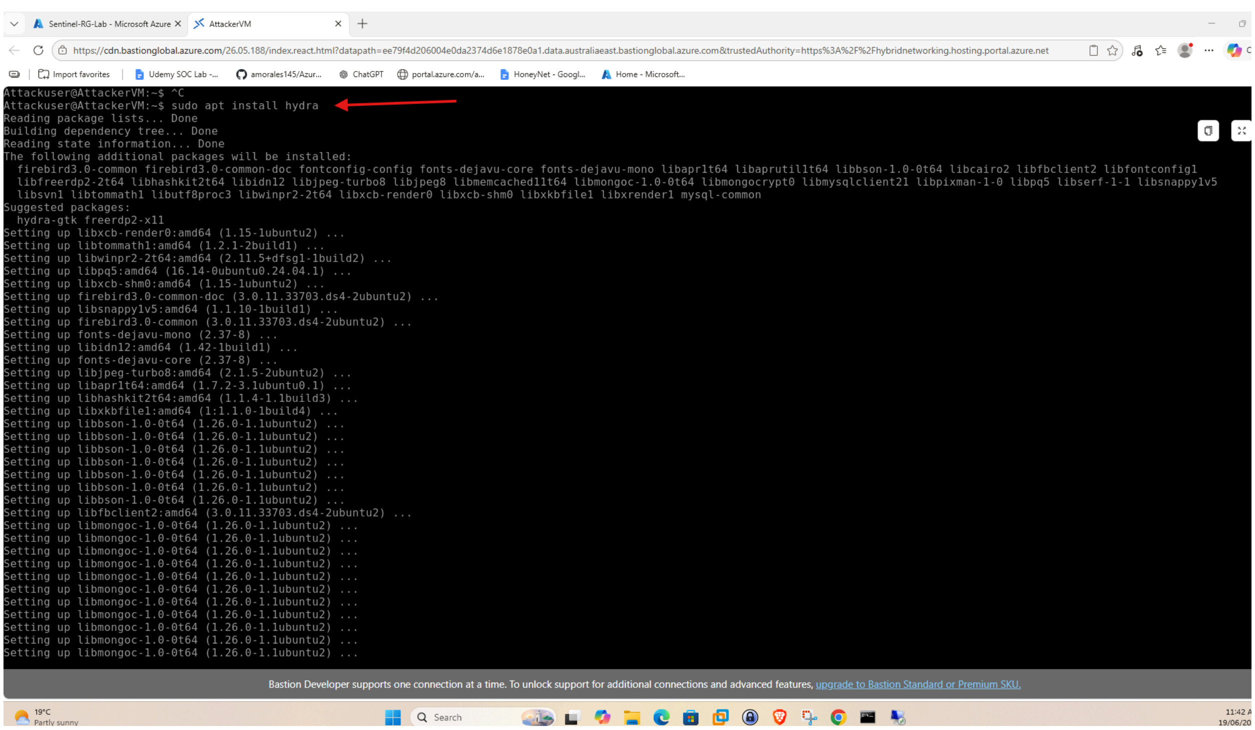


Connect to Attacker VM to install hydra for brute force:



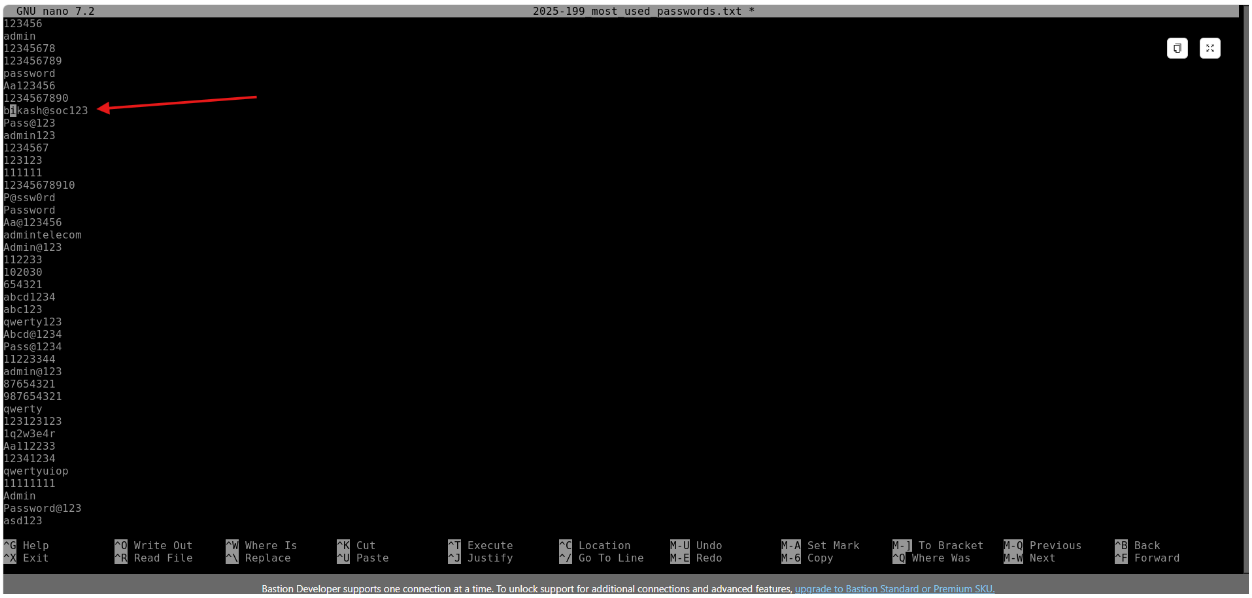
Update the VM and install Hydra
sudo apt update


Sentinel-RG-Lab - Microsoft Azure
AttackerVM
https://cdn.bastionglobal.azure.com/26.05.188/index.react.html?datapath=ee79f4d206004e0da2374d6e1878e0a1.data.australiaeast.bastionglobal.azure.com&trustedAu
Import favorites
Udemy SOC Lab -...
amorales145/Azur...
ChatGPT
portal.azure.com/a...
HoneyNet - Googl...
Home - Microsoft...
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
Attackuser@AttackerVM:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1031 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [259 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.4 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1656 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [326 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.8 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1110 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [251 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [40.4 kB]
Get:24 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [9972 B]
Get:25 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]


Sentinel-RG-Lab - Microsoft Azure
AttackerVM
https://cdn.bastionglobal.azure.com/26.05.188/index.react.html?datapath=ee79f4d206004e0da2374d6e1878e0a1.data.australiaeast.bastionglobal.azure.com&trustedAuthority=https%3A%2F%2Fhybridnetworking.hosting.portal.azure.net
Import favorites
Udemy SOC Lab -...
amorales145/Azur...
ChatGPT
portal.azure.com/a...
HoneyNet - Googl...
Home - Microsoft...
Attackuser@AttackerVM:~$ ^C
Attackuser@AttackerVM:~$ sudo apt install hydra
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
firebird3.0-common firebird3.0-common-doc fontconfig-config fonts-dejavu-core fonts-dejavu-mono libapr1t64 libaprutil1t64 libbson-1.0-0t64 libcairo2 libfbclient2 libfontconfig1
libfreerdp2-2t64 libhashkit2t64 libidn12 libjpeg-turbo8 libjpeg8 libmemcached11t64 libmongoc-1.0-0t64 libmongocrypt0 libmysqlclient21 libpixman-1-0 libpq5 libserf-1-1 libsnappy1v5
libsvn1 libtommath1 libutf8proc3 libwinpr2-2t64 libxcb-render0 libxcb-shm0 libxkbfile1 libxrender1 mysql-common
Suggested packages:
hydra-gtk freerdp2-x11
Setting up libxcb-render0:amd64 (1.15-1ubuntu2) ...
Setting up libtommath1:amd64 (1.2.1-2build1) ...
Setting up libwinpr2-2t64:amd64 (2.11.5+dfsg1-1build2) ...
Setting up libpq5:amd64 (16.14-0ubuntu0.24.04.1) ...
Setting up libxcb-shm0:amd64 (1.15-1ubuntu2) ...
Setting up firebird3.0-common-doc (3.0.11.33703.ds4-2ubuntu2) ...
Setting up libsnappy1v5:amd64 (1.1.10-1build1) ...
Setting up firebird3.0-common (3.0.11.33703.ds4-2ubuntu2) ...
Setting up fonts-dejavu-mono (2.37-8) ...
Setting up libidn12:amd64 (1.42-1build1) ...
Setting up fonts-dejavu-core (2.37-8) ...
Setting up libjpeg-turbo8:amd64 (2.1.5-2ubuntu2) ...
Setting up libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Setting up libhashkit2t64:amd64 (1.1.4-1.1build3) ...
Setting up libxkbfile1:amd64 (1:1.1.0-1build4) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libbson-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libfbclient2:amd64 (3.0.11.33703.ds4-2ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Setting up libmongoc-1.0-0t64 (1.26.0-1.1ubuntu2) ...
Bastion Developer supports one connection at a time. To unlock support for additional connections and advanced features, upgrade to Bastion Standard or Premium SKU.
19°C
Partly sunny
Search

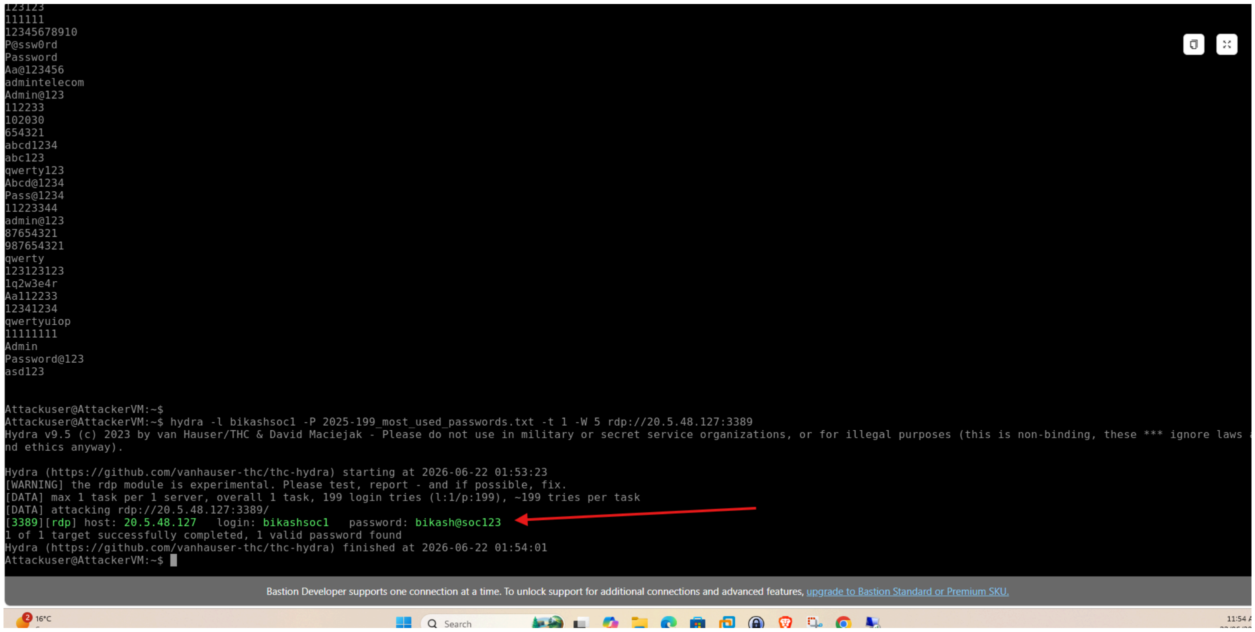
Create a password list or download from github the most commonly used password using wget
raw.githubusercontent.com/danielmiessler/SecLists/refs/heads/master/Passwords/Common-Credentials/2025-199_most_used_passwords.txt

Add the correct password to the text file and save it to simulate successful brute force attack
Nano 2025-199_most_used_passwords.txt



Attack the windows server VM using public IP rdp
Attackuser@AttackerVM:~$ hydra -l bikashsoc1 -P 2025-199_most_used_passwords.txt -t 1 -W 5 rdp://20.5.48.127:3389^

Successful logged in

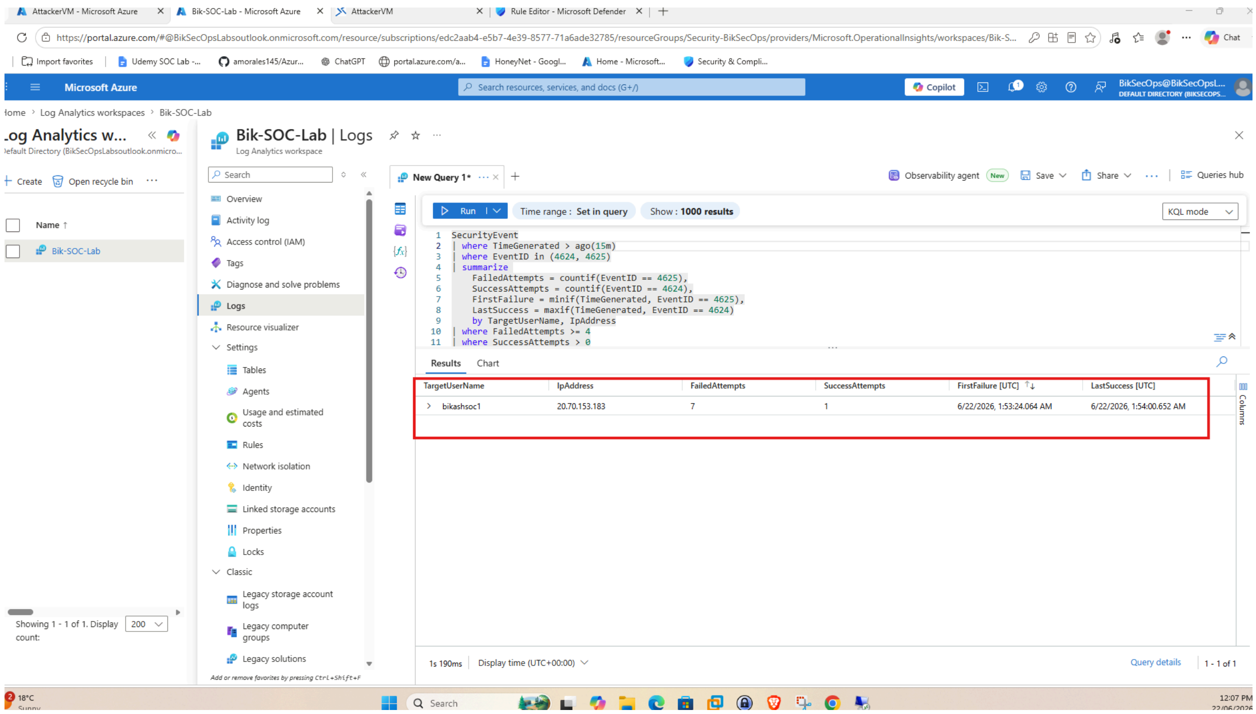


Confirm and Check the sentinel Logs for SecurityEvents are able to fetch:

Run the query to check the brute force attack followed by successful login:

```
SecurityEvent
| where TimeGenerated > ago(15m)
| where EventID in (4624, 4625)
| summarize
    FailedAttempts = countif(EventID == 4625),
    SuccessAttempts = countif(EventID == 4624),
    FirstFailure = minif(TimeGenerated, EventID == 4625),
    LastSuccess = maxif(TimeGenerated, EventID == 4624)
    by TargetUserName, IpAddress
| where FailedAttempts >= 7
| where SuccessAttempts > 0
| where LastSuccess > FirstFailure
```

This query analyzes Windows Security logs from the past hour to catch a brute force attack that ended in a successful login. It begins by pulling data exclusively from the `SecurityEvent` table and filtering for Windows Event IDs `4625` (failed login) and `4624` (successful login). Next, the `summarize` operator groups these events by unique pairs of `TargetUserName` and `IpAddress` while calculating the total count of failures and successes, alongside the timestamps for the very first failure and the latest success. Finally, the query applies strict logic gates to filter the results: it ensures there were at least four failed login attempts, that at least one login was successful, and that the successful login occurred *after* the failures had already begun. This effectively isolates instances where an attacker successfully guessed a user's password after multiple attempts from the same machine.



Create Detection Rule:

Name

Successful Brute Force Attempt

Description

Detects multiple failed logon attempts followed by a successful logon from the same IP address. This may indicate a brute-force attack, password spraying activity, or successful account compromise.

Severity

High

MITRE ATT&CK

Tactic

- Credential Access
- Initial Access

Techniques

- T1110 - Brute Force
- T1110.001 - Password Guessing
- T1110.003 - Password Spraying
- T1078 - Valid Accounts

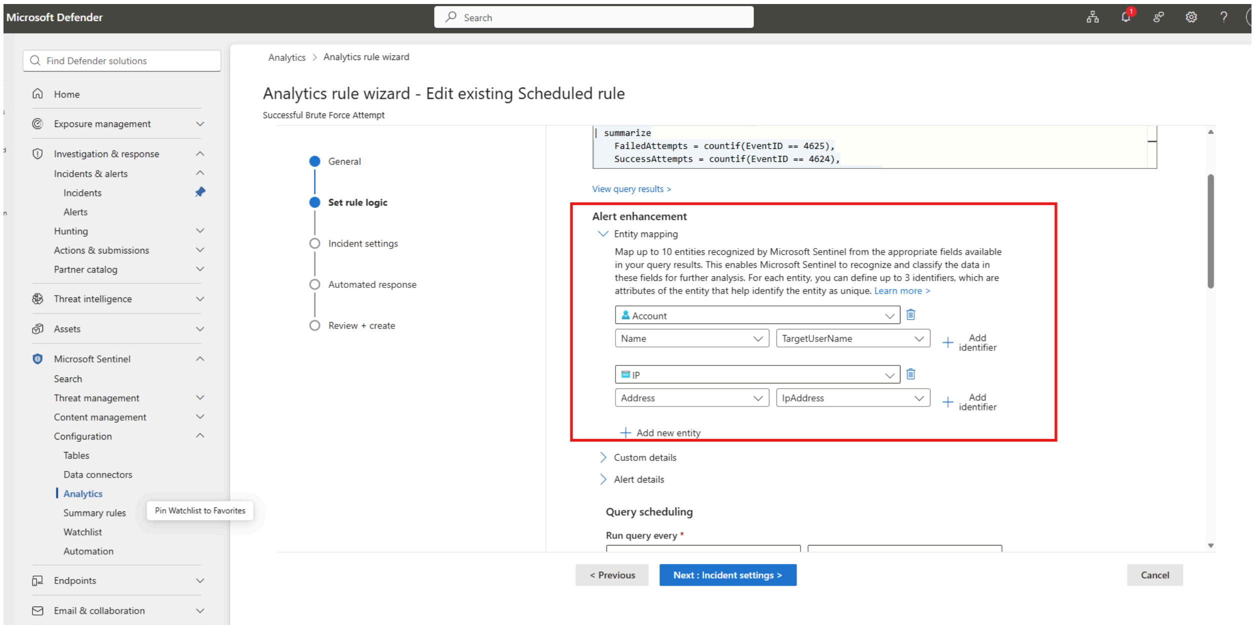
```
SecurityEvent

| where EventID in (4624, 4625)

| where LogonType in (2, 3, 10)

| summarize

   FailedAttempts = countif(EventID == 4625),

   SuccessAttempts = countif(EventID == 4624),

   LastFailed = maxif(TimeGenerated, EventID == 4625),

   SuccessTime = maxif(TimeGenerated, EventID == 4624)

   by TargetUserName, IpAddress

| where FailedAttempts >= 5

| where SuccessAttempts > 0

| where SuccessTime > LastFailed

| project TargetUserName, IpAddress, FailedAttempts, SuccessAttempts, LastFailed,
SuccessTime

| order by SuccessTime desc
```

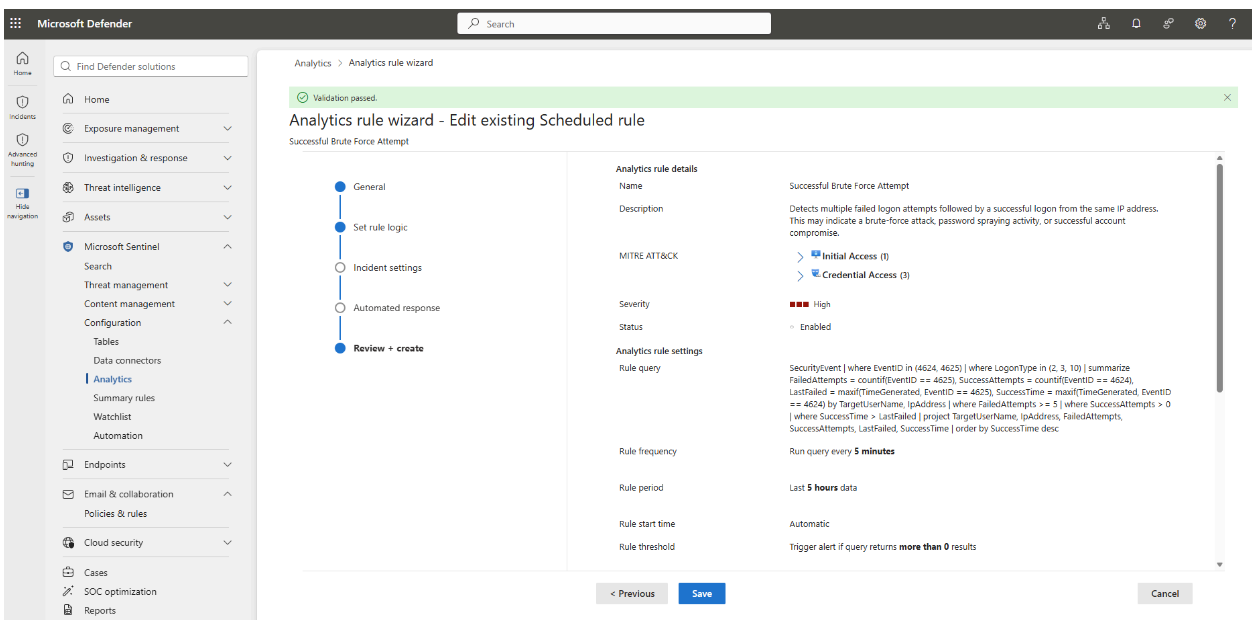
Entity Mapped:

Account

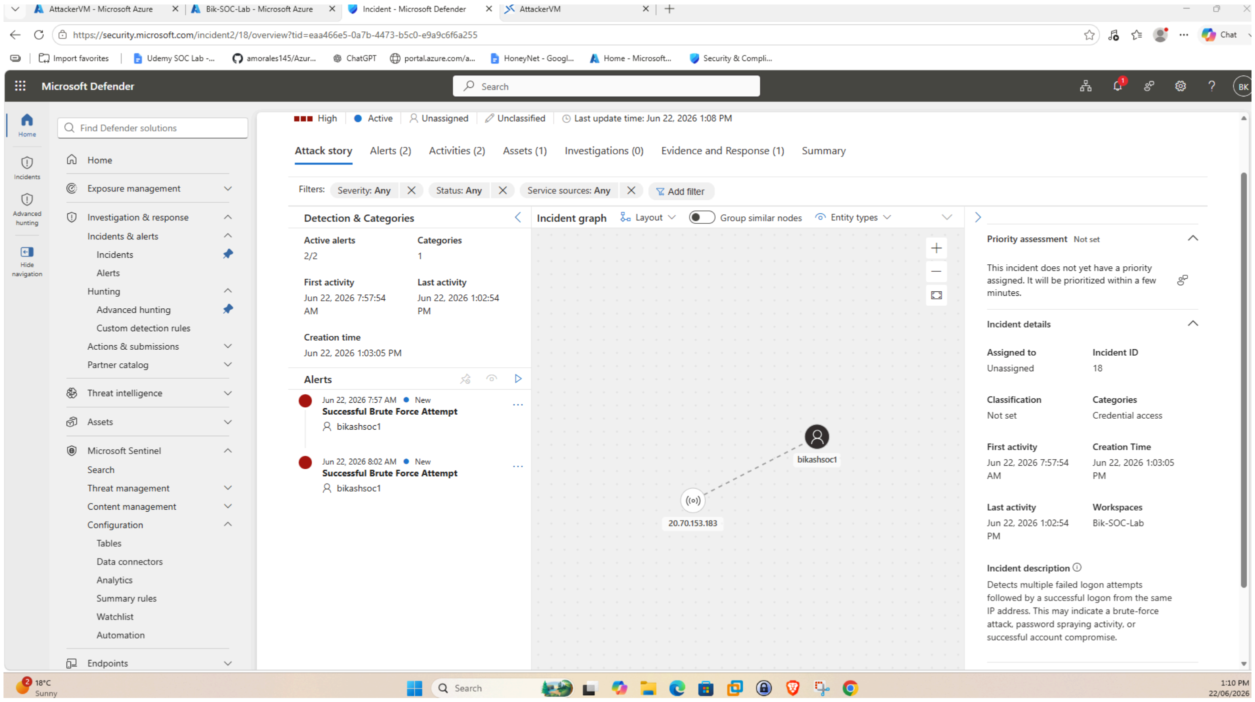
Name > TargetUserName

Address >IpAddress





Incident Has been triggered



Assigning and Creating the task based on playbook simulated:

Name

User / Host Review

Category

Investigate

Description

Review affected account and host associated with the alert.

Affected Account: bikashsoc1
Account Type: Privileged
Affected Host: Web01
Host Type: Production / External Facing

Validate whether the authentication activity is expected.
Review failed and successful logon events.
Identify source IP address and confirm attack path.
Determine whether unauthorized access occurred.

Recommended containment actions:

- Disable affected account

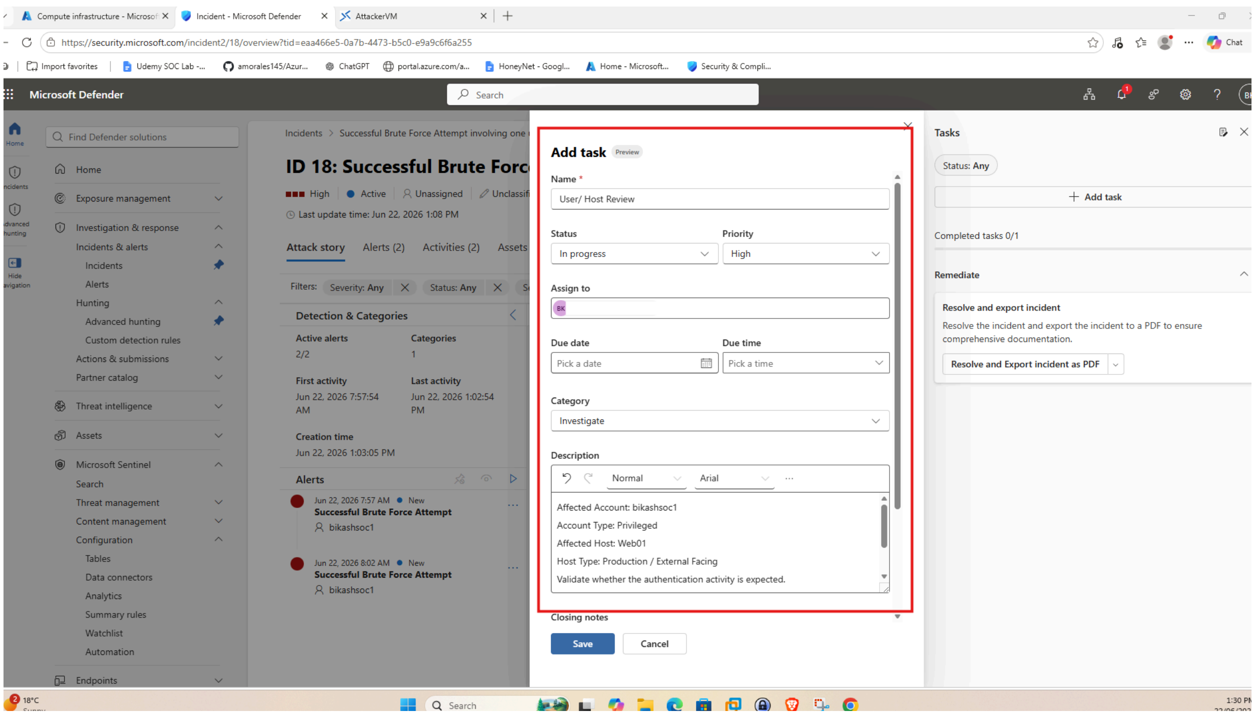
- Reset account password

- Revoke active sessions

- Block source IP address

- Isolate affected host if compromise is confirmed

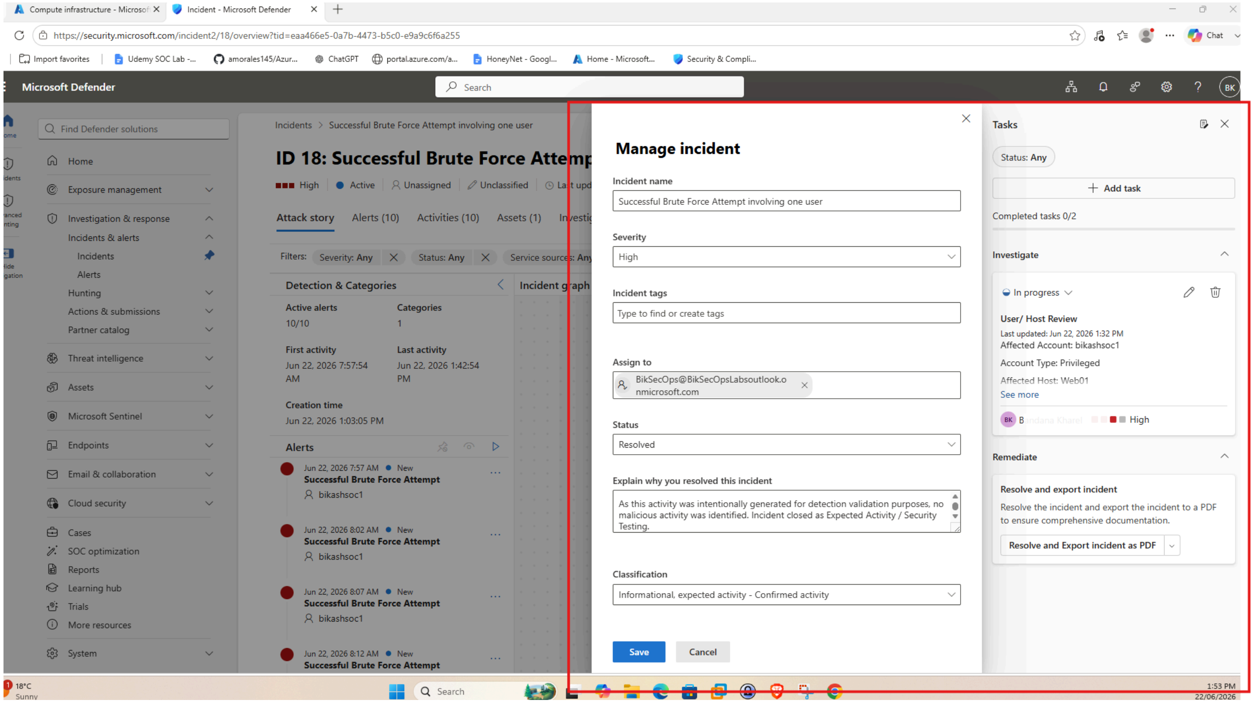
- Escalate to DFIR team if malicious activity is validated



Investigation completed. Multiple failed logon attempts followed by a successful authentication were observed and validated as part of authorized security testing within the lab environment.

In a real-world SOC environment, this incident would be escalated to the Incident Response (IR) or Digital Forensics and Incident Response (DFIR) team for further investigation, containment, and remediation activities.

As this activity was intentionally generated for detection validation purposes, no malicious activity was identified. Incident closed as Expected Activity / Security Testing.


Microsoft Defender
Manage incident
Incident name
Successful Brute Force Attempt involving one user
Severity
High
Incident tags
Type to find or create tags
Assign to
BikSecOps@BikSecOpsLabsoutlook.onmicrosoft.com
Status
Resolved
Explain why you resolved this incident
As this activity was intentionally generated for detection validation purposes, no malicious activity was identified. Incident closed as Expected Activity / Security Testing.
Classification
Informational, expected activity - Confirmed activity
Save
Cancel
Tasks
Status: Any
Add task
Completed tasks 0/2
Investigate
In progress
User/ Host Review
Last updated: Jun 22, 2026 1:32 PM
Affected Account: bikashsoc1
Account Type: Privileged
Affected Host: Web01
See more
High
Remediate
Resolve and export incident
Resolve the incident and export the incident to a PDF to ensure comprehensive documentation.
Resolve and Export incident as PDF
ID 18: Successful Brute Force Attemp
Attack story
Alerts (10)
Activities (10)
Assets (1)
Detection & Categories
Active alerts
10/10
Categories
1
First activity
Jun 22, 2026 7:57:54 AM
Last activity
Jun 22, 2026 1:42:54 PM
Creation time
Jun 22, 2026 1:03:05 PM
Alerts
Successful Brute Force Attempt
bikashsoc1